

Math 438: Mathematical Modeling

Instructor: Benjamin Ridenhour

Fall, 2019

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Office Hours: W 2:30–4:00pm (or by appt.)

Office: Brink Hall 327

Web: <https://bit.ly/31TaxS3>

Class Hours: MWF 1:30–2:20pm

Class Room: TLC 044

Course Description

The world around us is filled with processes which, at first glance, may seem quite complex. These complex processes span many fields, such as biology, physics, engineering, economics, language, literature, music, etc. Mathematical modeling presents us with a way to understand and predict the behavior of many types of systems. This first course on mathematical modeling equips students to create models to explain phenomena, fit models to relevant data, interpret models and their assumptions, and finally apply models to future problems. The book used for this course highlights certain common, real-world systems that are of interest to many STEM students. Additionally, the book provides many computational examples that can be used for undergraduate projects.

The structure of the class is as follows: Mondays will be used for reading, and we will generally read a chapter per week. The Wednesday class period will be for discussion of the reading and lecture on the topics. Fridays will be a practicum day where we practice problems and other relevant issues, with homework being assigned at the end of the class period. Participation in discussion and activities will be required on both Wednesday and Fridays. Student assessment will be based on participation, completing homework, 2 exams, and a final project.

The final project will be worked on throughout the semester. Students will work individually on projects with both instructor and peer feedback. For the project, students will pick a topic of their choosing. Creation of the model, fitting of supporting data, and interpretation of the results will all be critical aspects. At the end of the semester, students will create a scientific poster of their project. Presentation of the poster will be done in class, and consist of a 10 minute explanation followed by a Q&A session.

Scheduling oddity: Dr. Ridenhour will be travelling 23–27 September, so there will be no class that week.

Required Materials

- *A first course in mathematical modeling*. 5th edn. by FR Giordano, WP Fox, and SB Horton. Brooks/Cole: Boston.
- Functioning installations of R (<https://cran.r-project.org>) and RStudio (<https://www.rstudio.com>).

Prerequisites/Corequisites

Prerequisites: Math 310 (Diff. Eq.), 330 (Lin. Alg.)

Preferred: CS 120 (Comp. Sci. I)

Topics

1. Introduction to R, RStudio, and Rmd
2. Difference equations
3. Constructing mathematical models
4. Model Fitting
5. Modeling via experimentation and simulation
6. Optimization
7. Graph theory, decision theory, and game theory
8. Continuous models and their optimization
9. Dimensional analysis

Course Objectives/Learning Outcomes

Successful students will:

1. Learn how to create models that appropriate for discrete or continuous systems.
2. Gain mastery of standard methods for optimization and model fitting.
3. Be able to analyze and interpret models of various types.
4. Learn numerical and computational techniques to approximate/estimate solutions to models.

Final Exam Date

20 December (Friday), 12:45–2:45pm

Grading Policy

A typical grading scale will be used (i.e., $\geq 93\% = A$, $(93,90\%] = A-$, $(90,87\%] = B+$, ...). I reserve the right to curve the scale dependent on overall class scores at the end of the semester. Any curve will only ever make it easier to obtain a certain letter grade. The grade will count the assessments using the following proportions:

- 10% of your grade will be determined by attendance/participation in class.
- 30% of your grade will be determined by completing a final project.
- 20% of your grade will be determined by completing homework.
- 40% of your grade will be determined by two exams.